

Yiunfan Hu

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EDUCATION

BSc. Mechanical Engineering at Stony Brook University University Scholar; Dean’s list (all semesters); GPA: 3.98; U4 Standing	Stony Brook, NY Jun 2024–May 2027 (expected)
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Completed Coursework: [MEC203] *Engineering Graphics and CAD*, [MEC220] *Practical Electronics*, [MEC225] *Fundamentals of Machining Practices*, [MEC260] *Engineering Statics*, [MEC262] *Engineering Dynamics*, [MEC301] *Thermodynamics*, [MEC310] *Introduction to Machine Design*, [MEC325] *Manufacturing Processes*, [MEC363] *Mechanics of Solids*, [MEC364] *Introduction to Fluid Mechanics*

SKILLS

CAD	Autodesk Inventor, Autodesk Fusion, Onshape, SolidWorks
Software	C, C#, HTML/CSS/JS, MATLAB, Python
Mechanical	DFM, Rapid Prototyping, 3D Printing, CNC Machining, Manual Machining, Laser Cutting, Plasma Cutting
Electrical	Arduino, Soldering

EXPERIENCE

Stony Brook Robotics Team – MATE ROV Competition <i>Float Team Lead</i> - Led a small team in designing and building of the float for the 2026 Competition - Developed the hydraulic system and contributed significantly to arm design on the main ROV	Stony Brook, NY Sept 2025–present
Reactive Collision UAV <i>Research member at Prof. William Stewart’s Soft Flyers Group</i> - Assisted design a fixed-wing quadrotor resistant to collisions. - Fabricated major components such as the wings using a CNC router.	Stony Brook, NY Jun 2025–present
Shape Memory Alloy (SMA) Springs <i>Research member in a joint project with Interacting Robotic Systems Laboratory, Soft Flyers Group, and Yao Research Group, started in the summer before my first year</i> - Developed 2-DOF joint with Arduino control system for poster presentation in August 2024. - Designed and fabricated a specialized testing system to control the heating of SMA springs and detect the displacement and temperature response. - Experimentally validated theoretical models from literature and further expanded pre-established single-spring models to novel spring networks.	Stony Brook, NY May 2024–present

PROFESSIONAL TRAINING

NASA L’SPACE Proposal Writing and Evaluation Experience Academy <i>Participant – Team Role: Engineer</i> - Workforce preparation academy dedicated in writing and evaluating professional technical proposals taught by NASA scientists, engineers, and managers. - Developed a proposal focused on the laser ignition system for a rotating detonation rocket engine. - Served on a review panel for NASA solicitation responses and gained hands-on experience in evaluating proposals based on solicitation rubric and identifying strengths and weaknesses.	Virtual May–Aug 2025
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EXTRACURRICULAR PROJECTS

Combat Robotics at National Havoc Robot League (NHRL) <i>Team Captain</i> - Leading a team building combat robots in the 12lb weight class at NHRL. - Obtained rich experience in system integration, control systems, and metal manufacturing. - Developing a new robot with a novel wheel-less omni-directional drive system. - Competed in September 2023 and June 2024.	Norwalk, CT Jun 2023–present
Very Compact Electric Scooter <i>Independent Maker</i> - Designed and home-built one of the most compact electric scooters. - Gained practical experience in packaging and maximizing space utilization in design. - Successfully conducted road testing, followed up with upgrades, currently in-use for daily commute.	Stony Brook, NY Oct 2024–present